

No. of Functional Features Included in the Solution

Date	02 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Dataset Collection	Load the credit card application and credit record datasets for model development.	Data Collection	Done	High
2	Data Preprocessing	Remove duplicates, handle missing values, merge datasets, and encode categorical variables.	Data Preprocessing	Done	High
3	Machine Learning Model	Train and evaluate the Logistic Regression model for credit card approval prediction.	Model Building	Done	High
4	Prediction System	Predict whether a credit card application is approved or rejected based on user inputs.	Prediction Module	Done	High
5	Web User Interface	Provide a simple interface for users to enter applicant details and view prediction results.	Frontend (HTML/CSS)	Done	Medium
6	Flask Backend	Handle user requests, preprocess input data, load the trained model, and return predictions.	Backend (Flask)	Done	High
7	GitHub Repository	Maintain the complete source code using version control.	Version Control	Done	Medium
8	Render Deployment	Deploy the application online for real-time access through a web browser.	Cloud Deployment	Done	Medium

Feature Summary:

Metric	Count / Value
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Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6

Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	User input form, prediction result page	HTML forms, Result page
2	Backend / Logic	Data preprocessing, model prediction, Flask routing	Flask, Logistic Regression, Prediction logic
3	Database / Storage	Dataset and trained model storage	application_record.csv, credit_record.csv, model.pkl, encoders.pkl, columns.pkl
4	API / Integration	Flask request handling between frontend and ML model	Flask routes (app.py)
5	Security / Authentication	Basic input validation (no login required in this prototype)	Form validation